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Hamza Mohammed Hassanain

Software Engineer

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EDUCATION

B.Sc. Computer and Information Sciences, *Suez Canal University*
AWS Certified Solutions Architect – Associate, *Amazon Web Services*

2022 – 2026
credly.com/verify

EXPERIENCE

Software Engineer & Infrastructure Engineer

Nov 2025 – Present

Repovive

Part Time – United States, Remote

A competitive-programming platform: distributed judging, real-time contest tooling, and AI-assisted problem authoring.

- Led end-to-end design of the distributed judge (judge1 on **Judge0**) for sandboxed, multi-language auto-grading.
- Extended submission capacity **10x (50 to 500+ concurrent)** via a distributed, asynchronous execution pipeline.
- Cut infrastructure cost **95% (\$600 to \$25/mo)** by migrating resource-heavy tasks from serverless to Docker/Nginx/Redis.
- Isolated **20+ languages** in resource-limited Docker images; redesigned the DB layer for **50MB+** contest test cases.
- Built a real-time collaborative dev space using **WebSockets** and **Yjs CRDTs** over a Monaco live-syncing editor.
- Co-founded **Build v2**, an AI problem-authoring IDE: 28-tool agent pipeline with execution locks and atomic cancellation.
- Sustained 10-minute streaming agent sessions, scaled server capacity **10x** for test runs, and mentored junior engineers.
- Tech: TypeScript, Next.js, AWS, Docker, Nginx, Redis, MongoDB, Yjs.

Problem Setter & Instructor

Jul 2023 – Jul 2025

ICPC SCU Community

Remote

- Authored competitive-programming problems for **ECPC** qualifiers, balancing difficulty alongside the committee.
- Mentored **100+ trainees** and organized regular programming contests for the university community.

OPEN SOURCE CONTRIBUTIONS

- **GitLab Go SDK (client-go)**: typed group integrations – Slack, Jira, MS Teams, Mattermost, Harbor, +3 | [!2692](#), [!2691](#), [!2675](#), [!2670](#).
- **GitLab Terraform provider**: new Harbor & Mattermost integration resources + group SAML attribute | [!2881](#), [!2909](#), [!2852](#).
- **GitLab monorepo**: `Json.safe_parse` hardening and Vue lint compliance | [!221163](#), [!221158](#).
- **LLVM / Clang**: `constexpr X86 cvtpd2ps`, `AArch64/ARM CLS` → `ISD`: `:CTLS`, `LLDB SWIG` tests | [#169980](#), [#178885](#), [#178430](#), [#173473](#).
- **Apple Swift NIO**: idempotent `NIOFileSystem` directory creation | [#3410](#).

PROJECTS

Autonomous AI Tutor & Curriculum Engine, Graduation Project

github.com/sigma-loop

- Led backend & DevOps for an 8-person team building an on-demand AI course, lesson, and challenge generator.
- Graded code in a Judge0 sandbox and math proofs via a confidence-scored LLM; owned all CI/CD and AWS deployments.

Competitive-Programming Judge – AWS Reference Architecture

github.com/HamzaHassanain/CP-Judge-AWS

- Designed an AWS reference architecture: Judge0 `isolate` on ECS-on-EC2, Fargate orchestrator, Spot ASG on queue depth.
- Automated post-round cheating detection with Step Functions, AWS Batch, Lambda/Bedrock, and Athena.

Polyman CLI

npmjs.com/package/polyman-cli | github.com/HamzaHassanain/polyman

- Built a TypeScript CLI automating the Codeforces Polygon workflow: test generation, validation, and direct API sync.

Systems in C++ – HALlocator & cpress

[HALlocator](#) | [cpress](#)

- Built a Red-Black-Tree best-fit memory allocator ($O(\log n)$) and an epoll-driven C++ web framework, both from scratch.

TECHNICAL WRITING

- Authored ["What Every Programmer Should Know About Memory"](#), a series on CPU caches, virtual memory, and optimization.
- Published ["I Built a Backend Web Framework from Scratch in C++"](#), covering non-blocking I/O, HTTP parsing, and thread pools.

ACHIEVEMENTS

ICPC ECPC & ACPC – Competitive Programming

icpc.global/YC7Z02WQCPIB

17th of 500 teams at ECPC (2024); two-time ACPC regional qualifier (2024, 2025), top 250 in the Africa & Arab region.

SKILLS

Cloud & DevOps

AWS, Docker, Nginx, Redis, Terraform, CI/CD (GitHub Actions), DigitalOcean, Linux

Backend & Full-Stack

TypeScript, Node.js, Express.js, Next.js, React, MongoDB, PostgreSQL, REST APIs, WebSockets, Yjs (CRDTs)

Languages & Systems

C++, Go, Rust, Python, SQL; System Design, Distributed Systems, Message Queues